Speech and Audio Processing

04 - speech synthesis

Shinnosuke Takamichi / 高道 慎之介

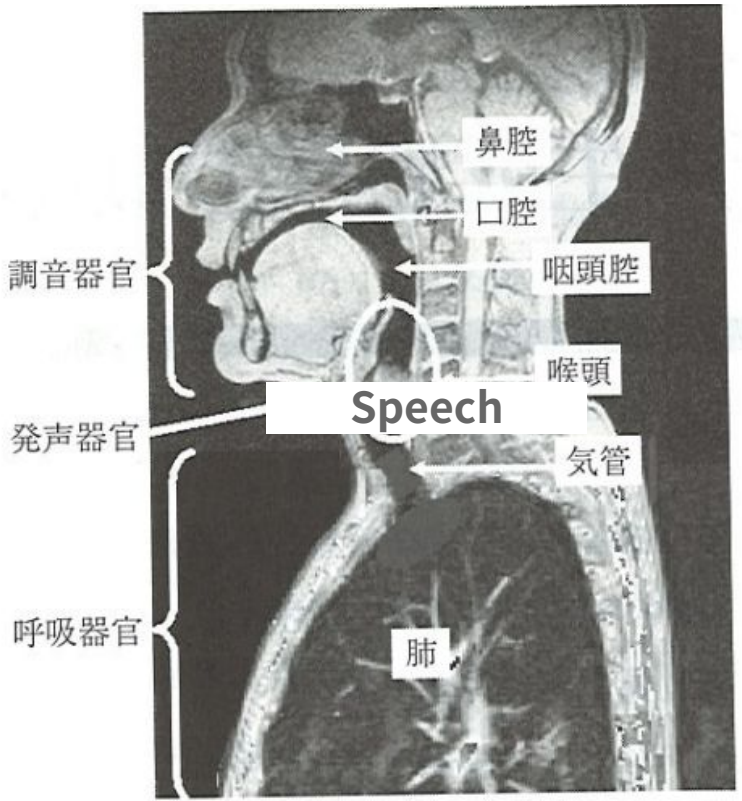
Takamichi Laboratory
慶應義塾大学 高道研究室

Class schedule (tentative)

01	Oct. 02	Introduction
02	Oct. 09	Speech production / 音声生成
03	Oct. 16	Speech analysis and deep learning / 音声分析と深層学習
04	**Oct. 23**	**Speech synthesis / 音声合成**
05	Oct. 30	Self-supervised learning / 自己教師あり学習
06	Nov. 6	**Guest Talk (acoustic signal processing) / ゲストトーク (音響信号処理)**
07	Nov. 13	Foundation model / 基盤モデル
08	Nov. 27	Advanced topics / 発展的話題
09	Dec. 04	**Guest Talk (music information processing) / ゲストトーク (音楽情報処理)**
10	Dec. 11	Speech system building / 音システム構築
11	Dec. 18	Speech system building / 音システム構築
12	Dec. 25	Speech system building / 音システム構築
13	Jan. 08	Speech system building / 音システム構築
14	Jan. 22	Final presentation / 最終発表

Last week's class

How is feature extracted? How are neural networks used to model?

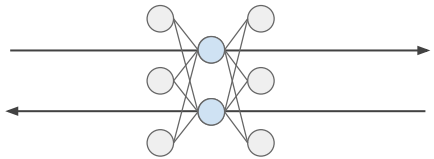


Analysis

Feature

Synthesis

Recognition



Label

Synthesis

Today's class: text-to-speech (TTS)

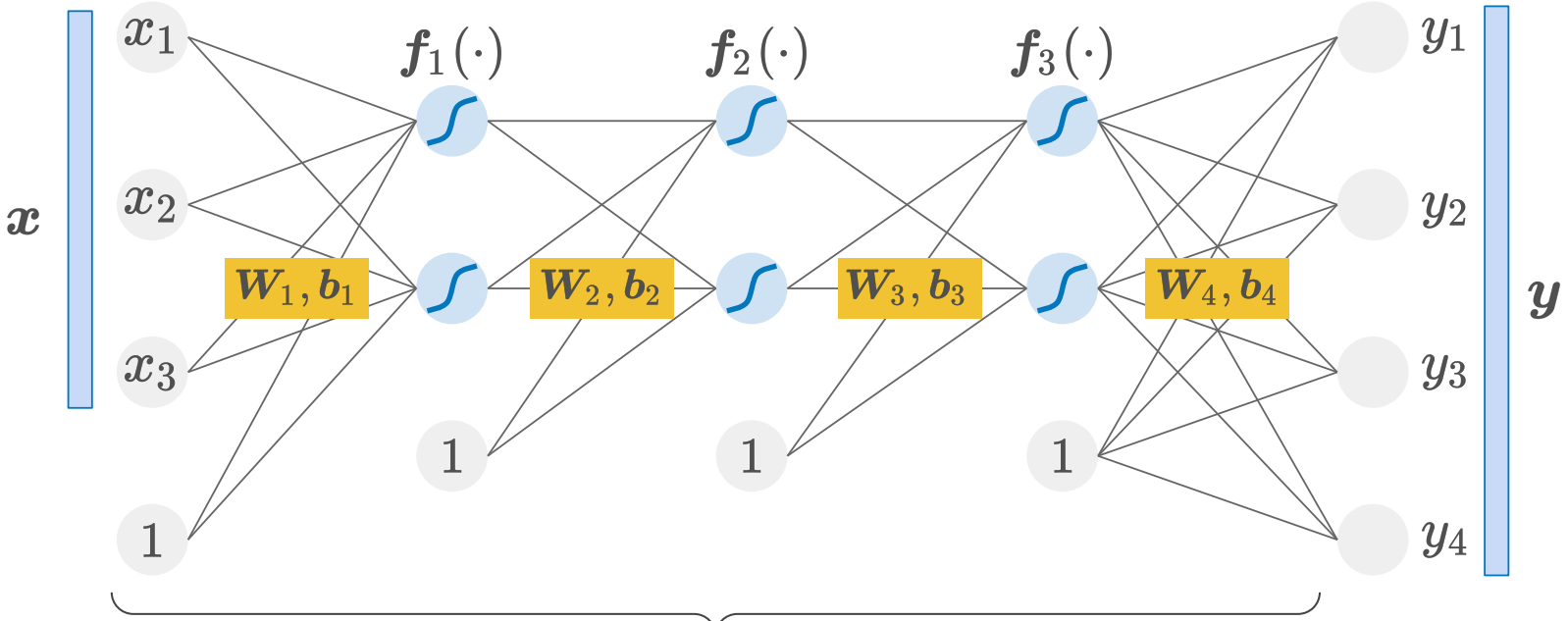
What happens in computers when artificially synthesizing speech



Review of last week’s content / 前回の復習

Neural network: multilayer perceptron (MLP)

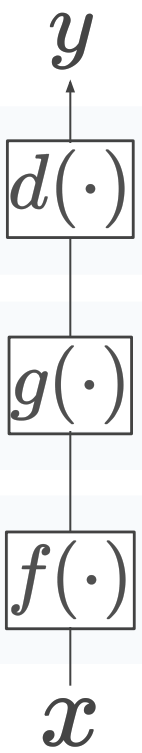
Stacking simple perceptrons enables "deep" non-linear transformations



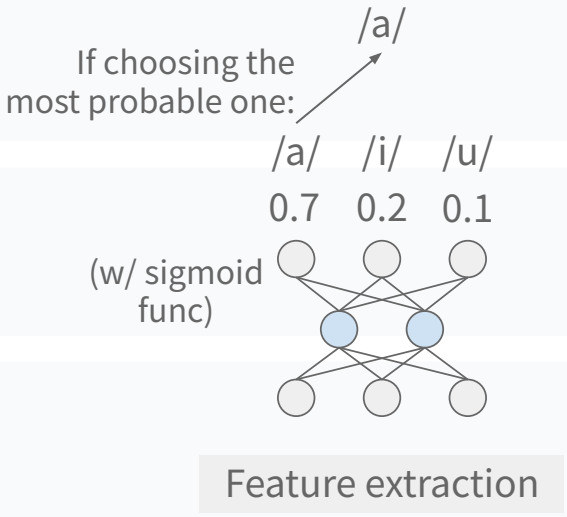
$$y = f_\theta(x)$$

Set of W_*, b_* (function with parameters θ)

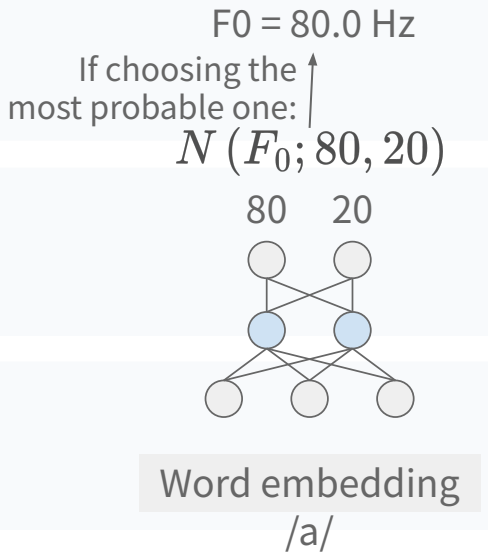
Example



Speech-to-text (cont. to disr.)

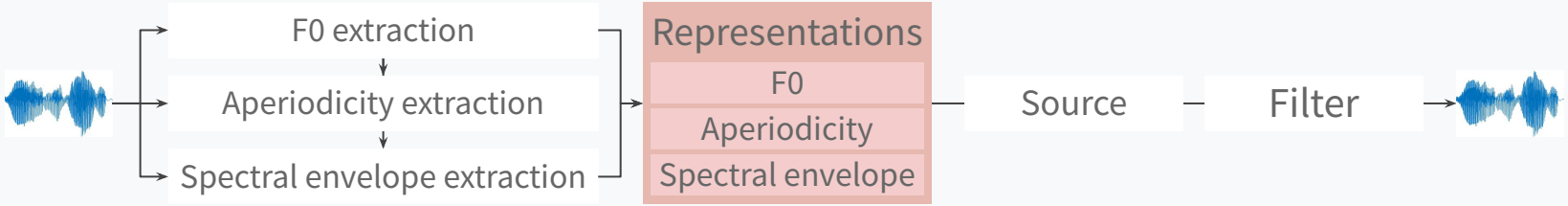


Text-to-speech (disr. to cont.)

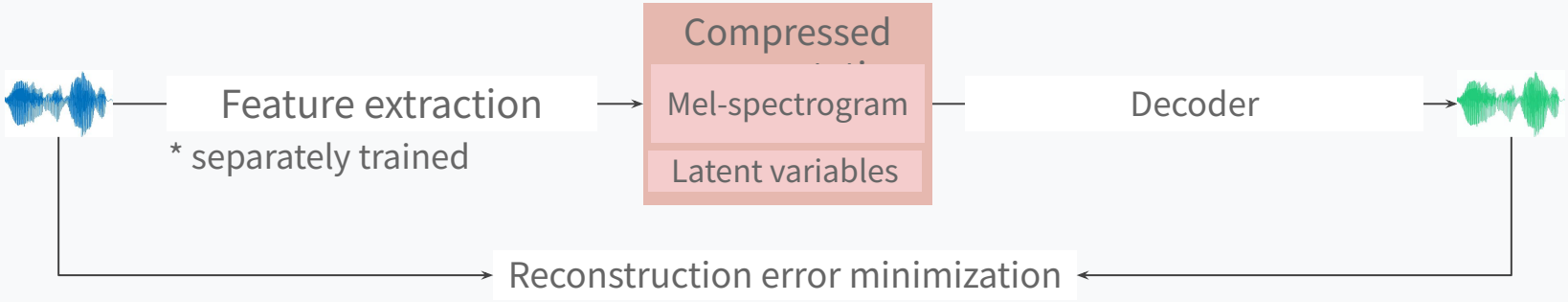


Neural vocoder

Signal processing-based vocoder



Neural vocoder



History / 歴史

Voder, the first synthesizer (1939)



Daisy Bell by world's first "singing computer" (1961)



Formant synthesis & DECtalk (1983)

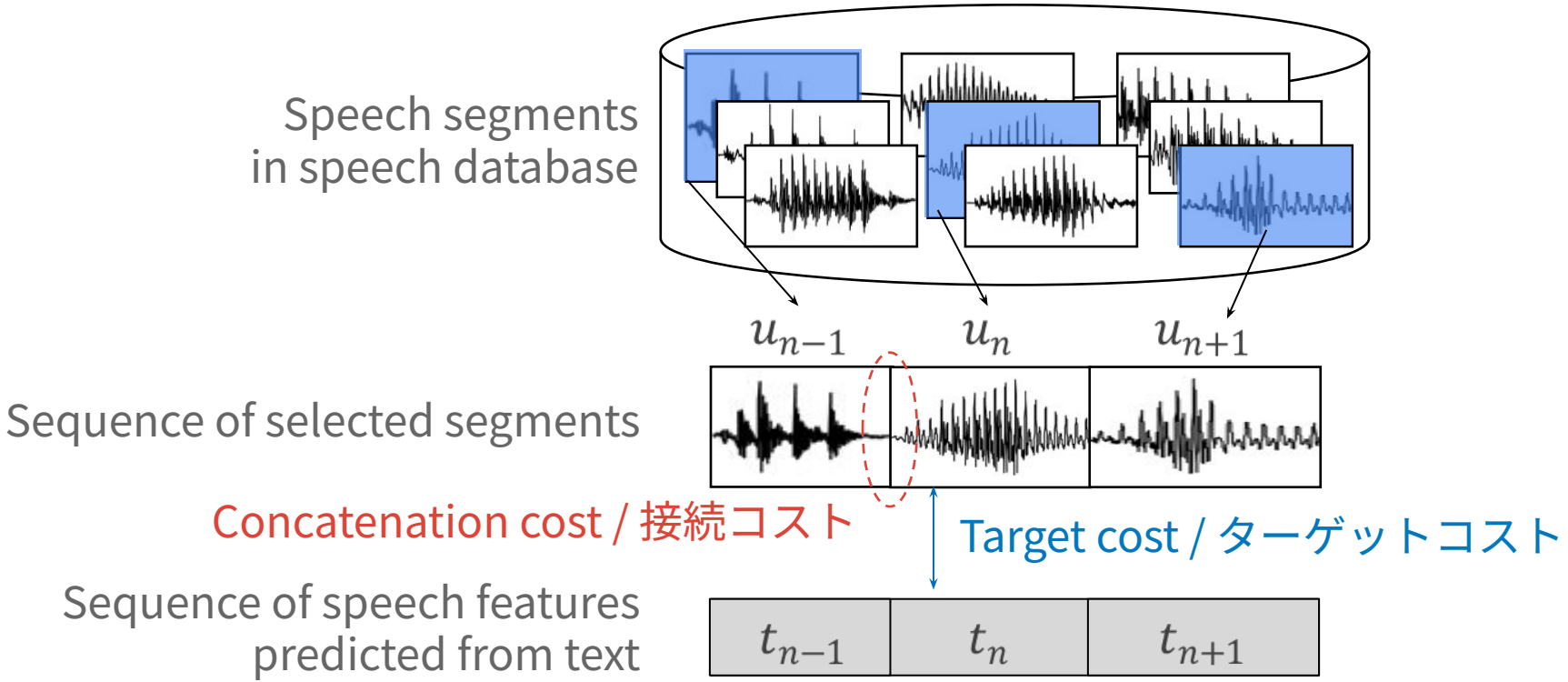


Formant synthesis still exists in Japanese TTS

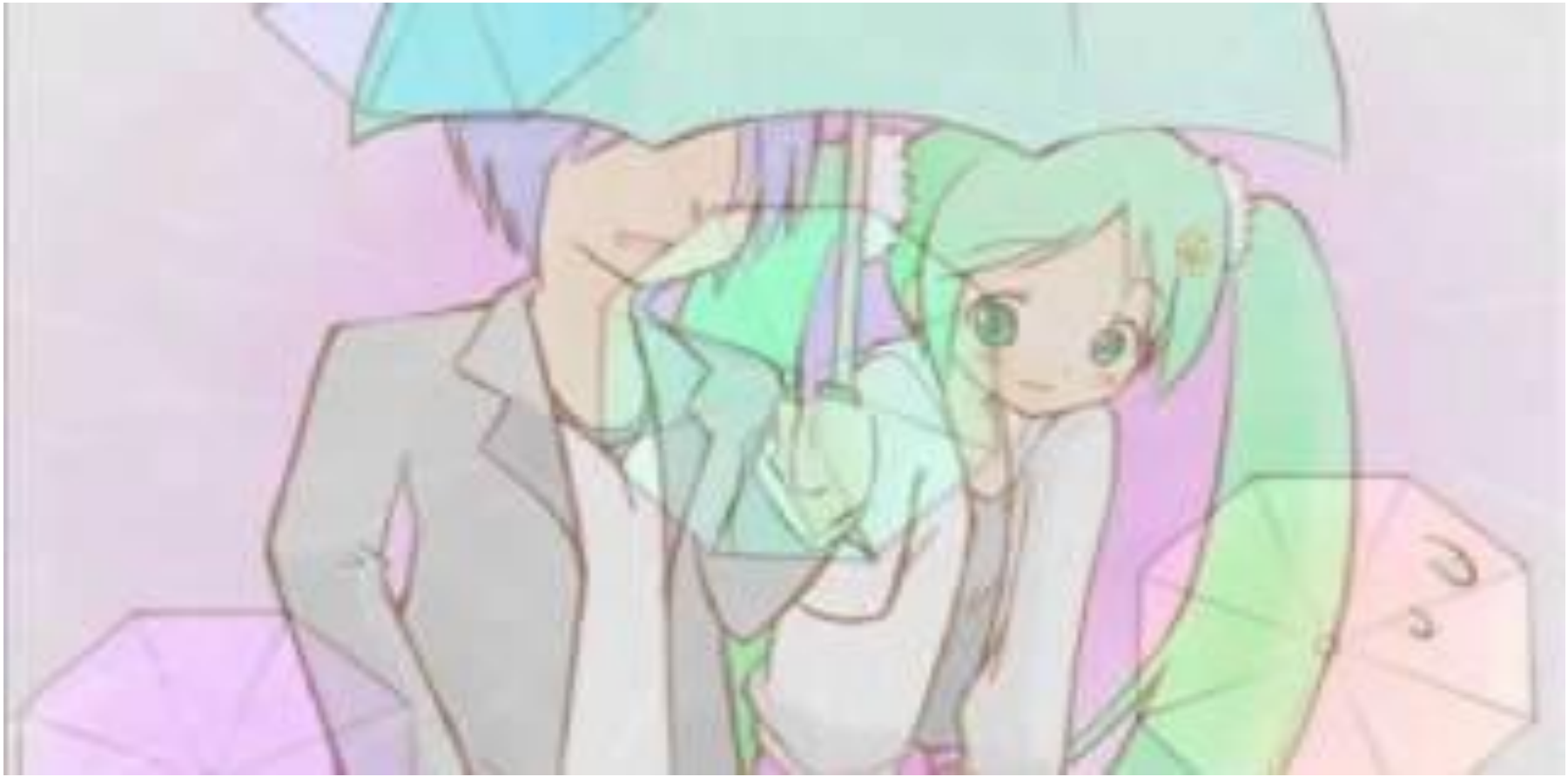


Era of data-driven synthesis: unit selection (素片選択) (1990s)

Selecting segments to minimize total costs including two costs.



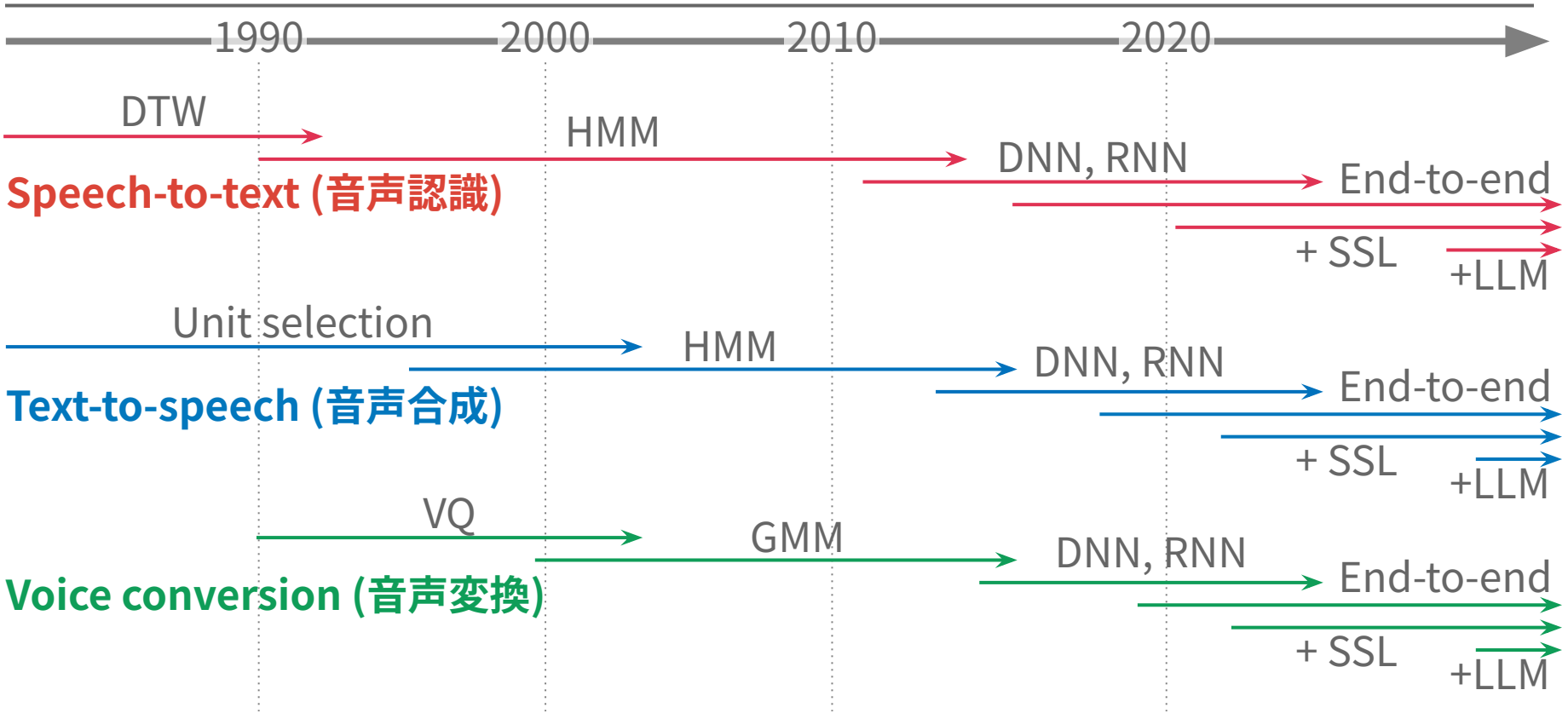
First generation of VOCALIOD (e.g., Hatsune Miku) is based on unit selection synthesis



ElevenLabs: one of the leading company in deep learning era

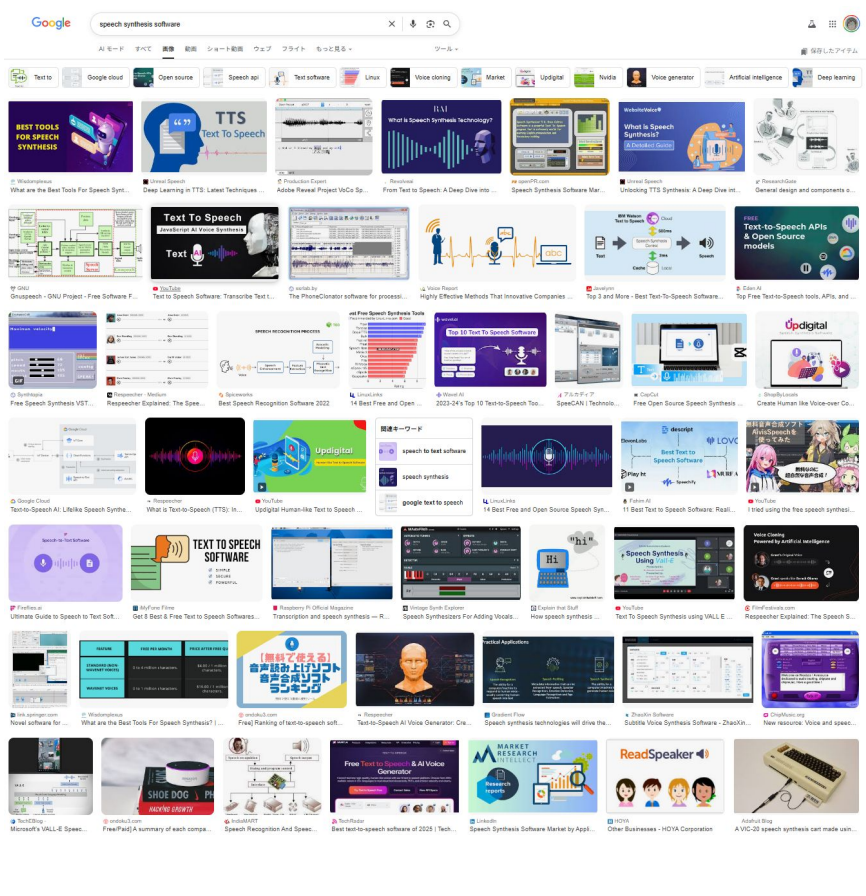


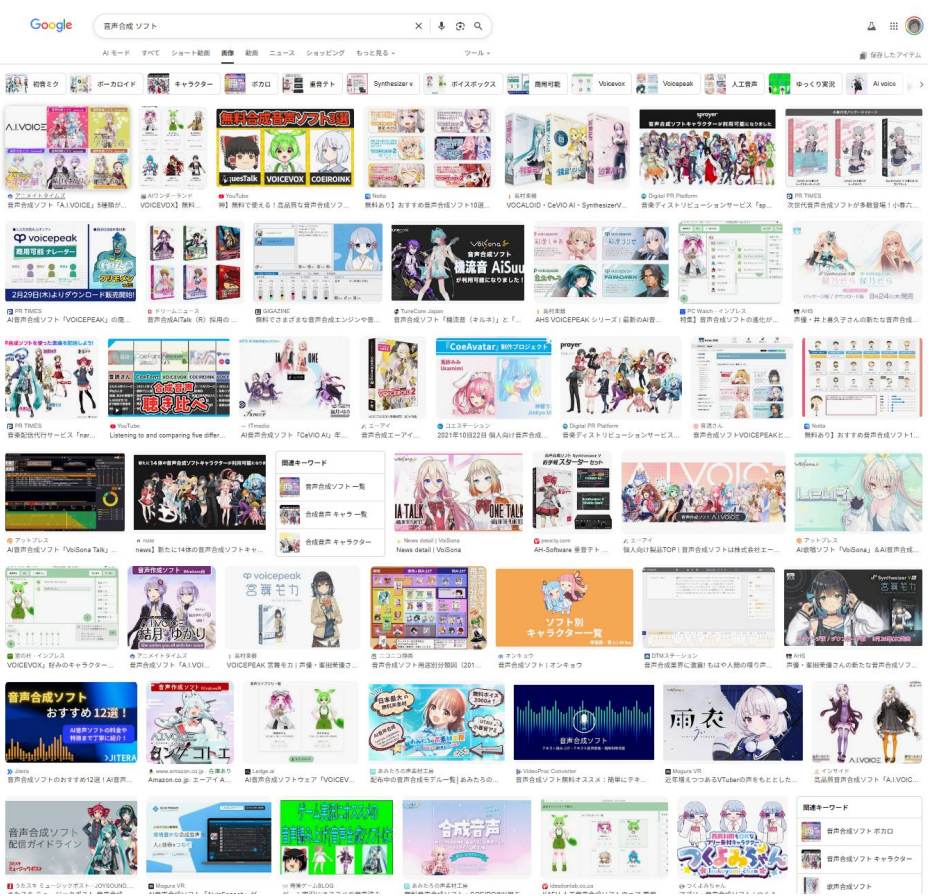
Data driven approach to TTS



* DTW: dynamic time warping, HMM: hidden Markov model, DNN: deep neural network, RNN: recurrent neural network
VQ: vector quantization, GMM: Gaussian mixture model, SSL: self-supervised learning, LLM: large language model

Nowadays

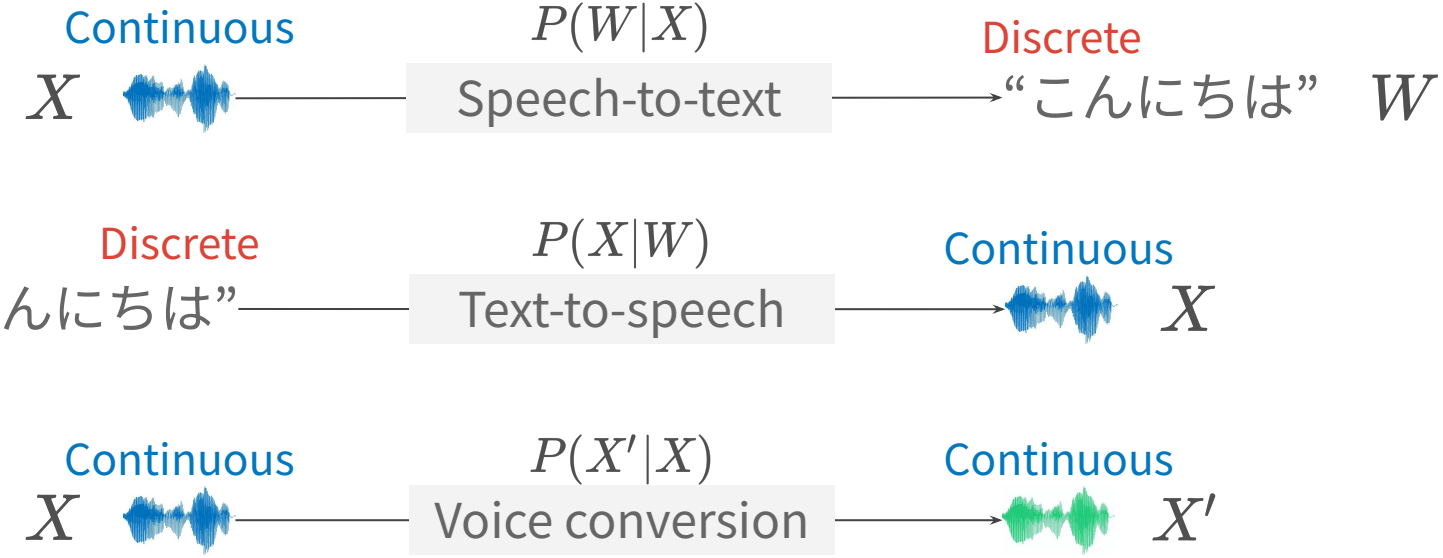

speech synthesis software
BEST TOOLS FOR SPEECH SYNTHESIS
TTS
Text To Speech
Text To Speech
JavaScript AI Voice Synthesis
Text
Text-to-Speech APIs & Open Source models
TEXT TO SPEECH SOFTWARE
Hi
Speech Synthesis Using Kalid
Updigital
speech to text software
speech synthesis
google text to speech
【無料で使える】音声読み上げソフト 音声合成ソフト ランキング
SHOE DOG
MARKET RESEARCH INTELLIGENCE
Research reports
ReadSpeaker


音声合成 ソフト
無料合成音声ソフト3選
VOICEVOX
COEIROINK
voicepeak
AiSuu
CoeAvatar
A.I.VOICE
IA TALK
ONE TALK
ソフト別キャラクター一覧
音声合成ソフト おすすめ12選！
AI音声ソフトの特徴をそれぞれ紹介！
雨衣
関連キーワード
音声合成ソフト ボカロ
音声合成ソフト キャラクター
歌声合成ソフト

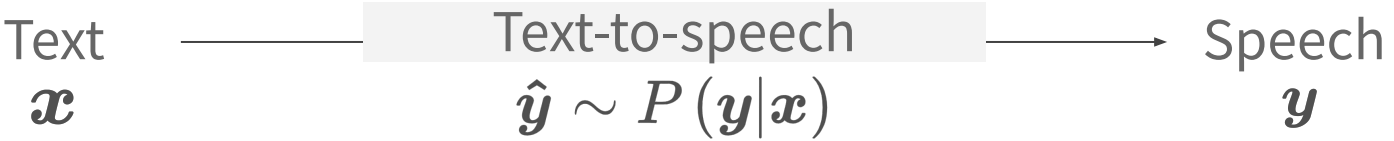
Mathematical formulation / 定式化

Formulation / 定式化

Variable-length conversion from a {continuous, discrete} sequence to another {continuous, discrete} sequence.



Formulation / 定式化



Why are statistical models used?

- Speech production is probabilistic
  - Motor command: It changes even if you pronounce the same word.
  - Radiation: Frequency phase varies when you move.

- Pronunciation of text is not deterministic
  - Homophone (同音異義語)：移動 (i do u) vs. 異同 (i do u)
  - Heterophone (同形異音語)： 市場 (shi jo u) vs. 市場 (i chi ba)

Mitigating into subproblems / 部分問題への緩和

$P(\boldsymbol{y}|\boldsymbol{x})$ Original formulation corresponding to “end-to-end” modeling.

$$= \int\int P(\boldsymbol{y}|\boldsymbol{o},\boldsymbol{l},\boldsymbol{x})P(\boldsymbol{o},\boldsymbol{l}|\boldsymbol{x})d\boldsymbol{o}d\boldsymbol{l}$$

Introducing intermediate features: speech feats *o*, and linguistic feats *l*.

$$= \int\int P(\boldsymbol{y}|\boldsymbol{o},\boldsymbol{l},\boldsymbol{x})P(\boldsymbol{o}|\boldsymbol{l},\boldsymbol{x})P(\boldsymbol{l}|\boldsymbol{x})d\boldsymbol{o}d\boldsymbol{l}$$

Approximating with point estimates (点推定)

$$\approx P(\boldsymbol{y}|\boldsymbol{o},\boldsymbol{l},\boldsymbol{x})P(\boldsymbol{o}|\boldsymbol{l},\boldsymbol{x})P(\boldsymbol{l}|\boldsymbol{x})$$

Assuming statistical independence of *y* on {*l*, *x*}, and of *o* on *l*.

$$= P(\boldsymbol{y}|\boldsymbol{o})P(\boldsymbol{o}|\boldsymbol{l})P(\boldsymbol{l}|\boldsymbol{x})$$

Mitigating into subproblems / 部分問題への緩和

$$P(\boldsymbol{y}|\boldsymbol{o})P(\boldsymbol{o}|\boldsymbol{l})P(\boldsymbol{l}|\boldsymbol{x})$$

Prob. of waveform given by speech feats. (vocoder / ボコーダ)

Prob. of speech feats. given by linguistic feats. (acoustic model / 音響モデル)

Prob. of linguistic feats. given by text (text analysis / テキスト解析)

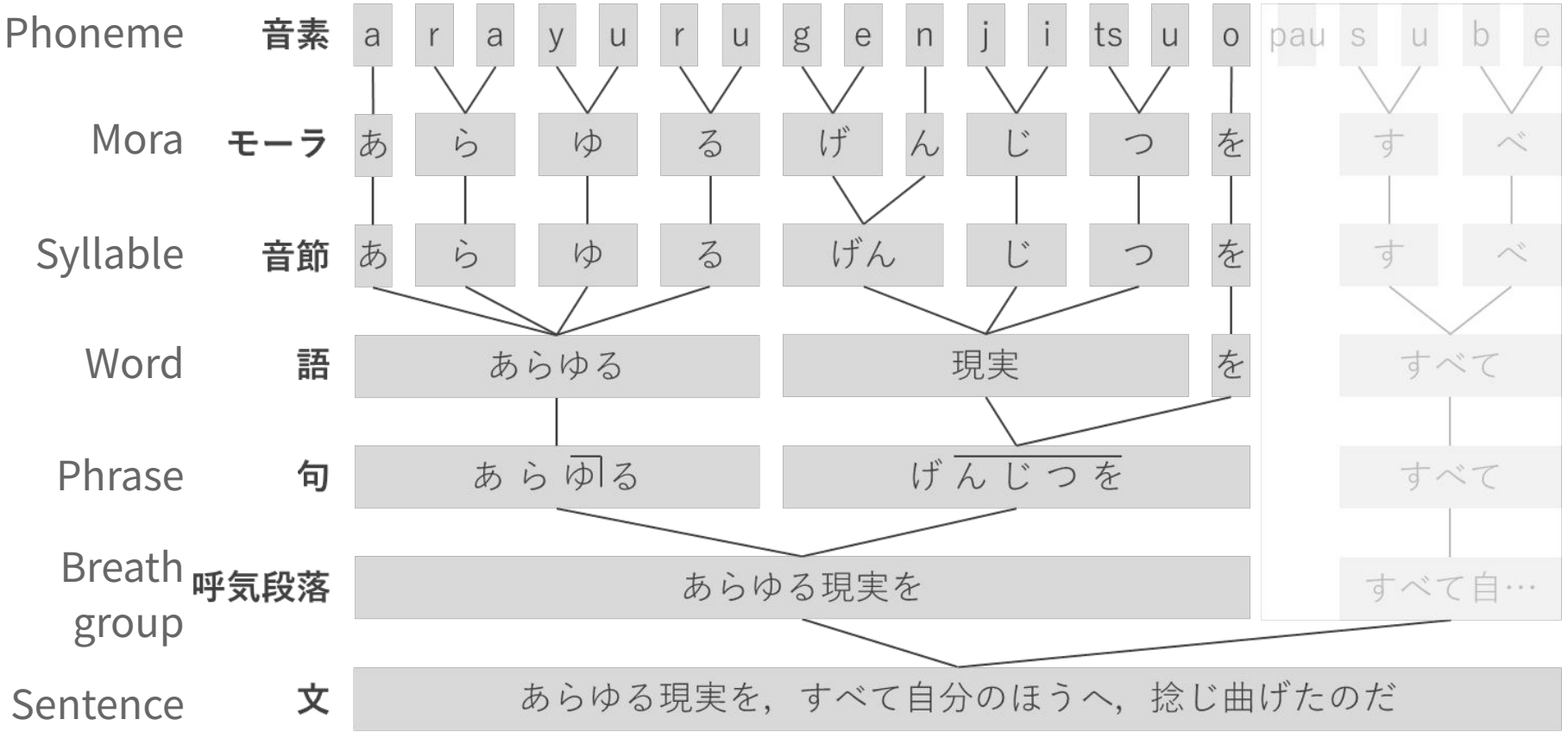
Handling waveform-level fluctuation, e.g., aperiodicity (非周期性)

Handling command-level or feat.-level fluctuation, e.g., movement variation.

Handling linguistic feat.-level fluctuation, e.g., accent, pronunciation.

Text analysis in Japanese / 日本語のテキスト解析

Linguistic units in Japanese / 日本語における言語単位



Phoneme / 音素

- Phoneme / 音素
  - Vowel (母音, V): /a, i, u, e, o/
  - Consonant (子音, C) /p, b, t, d, k, g, c, s, z, h, m, n, r, w, j, N, q/
- Mora / 拍
  - Smallest unit of timing.
  - “ー” (長音符), “っ” (促音), “ん” (撥音) are counted as a unique mora
- Syllable / 音節
  - Basic unit defined as a vowel with optional consonants.

語 Word	音節 Syllable	モーラ Mora
おはよう o-ha-yo-u	お／は／よ／う	お／は／よ／う
かんぬき ka-n-nu-ki	かん／ぬ／き	か／ん／ぬ／き
いっぱつ i-p-pa-tsu	いっ／ぱ／つ	い／っ／ぱ／つ

Isochrony / 等時性

Any languages can be divided into equal rhythmic positions.

- Japanese: mora-timed language / モーラ等時性言語

wa ta shi wa to sho ka n e i ki ma shi ta

わ た し は と しょ か ん へ い き ま し た。

♪ ♪ ♪ ♪ ♪ ♪ ♪ ♪ ♪ ♪ ♪ ♪ ♪ ♪

- Mandarin: syllable-timed language / シラブル等時性言語

我 去 图 书 馆

♪ ♪ ♪ ♪ ♪

- English: stress-timed language / ストレス等時性言語

I went to the library to study for the exam.

♪ ♪ ♪ ♪

Word and accent / 語，アクセント

- Word / 語
  - Linguistic unit consisting of 1+ morphemes (形態素)

- Accent / アクセント
  - Relative emphasis of a certain syllable (or mora)
  - **Japanese (Tokyo dialect) is a pitch-accent language**
    - Two level (**L**ow or **H**igh) accent.
    - Pitch-fall positions changes the meaning of words.
    - *N*-mora words have N+1 kinds of accents.
      - type 0 (no pitch-fall): LHH(H)
      - type 1: HLL(L)
      - type 2: HHL(L)

Accent table / アクセント表

“\” indicates pitch-fall positions. “¯” indicates no pitch-fall positions.

表 2-3 日本語のアクセント。“＼”がアクセント核（下がる位置）、“¯”はアクセント核のないことを表します。

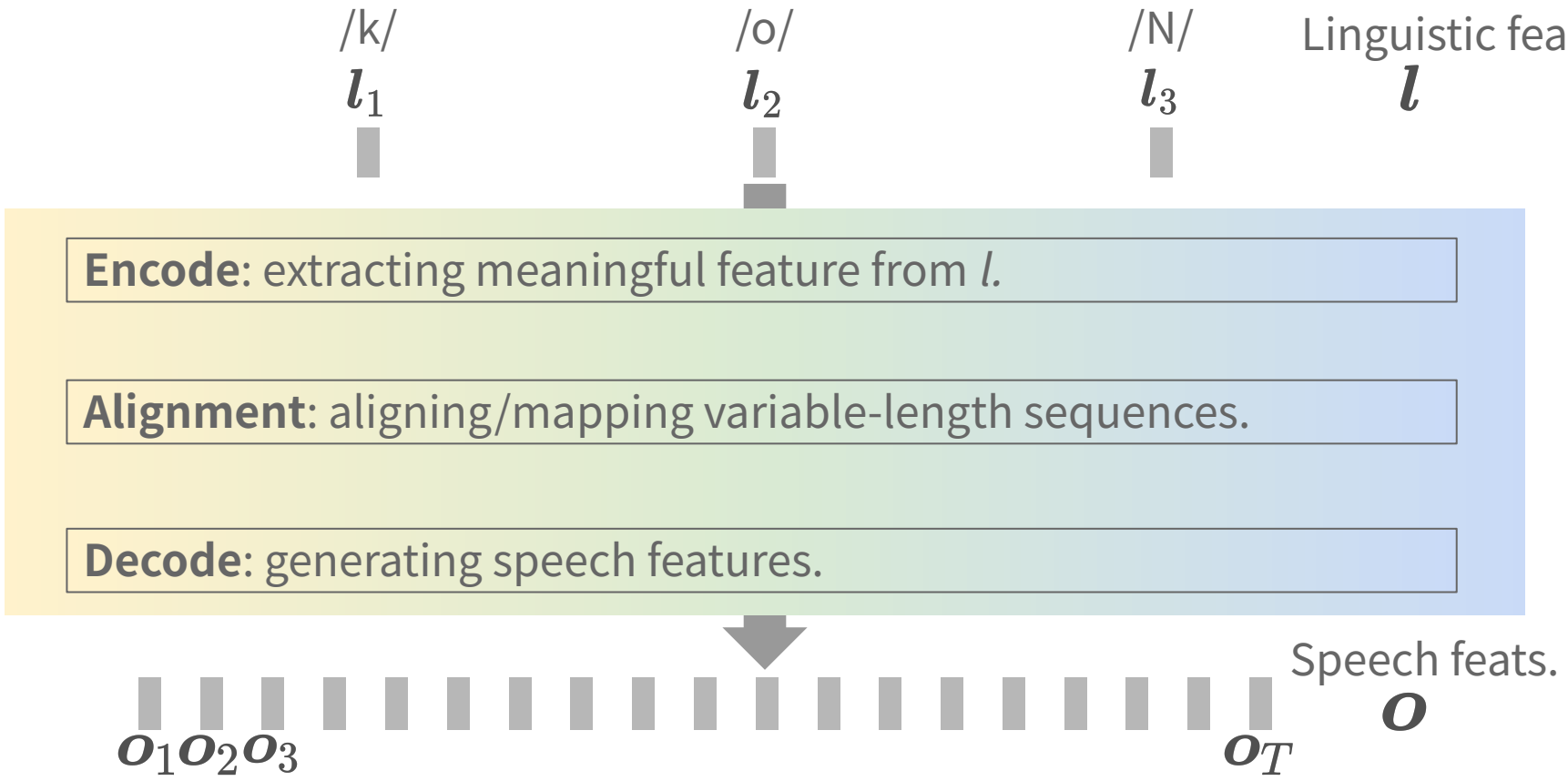
	1拍語 1 mora	2拍語 2 morae	3拍語 3 morae	4拍語 4 morae
0型 Type 0: no falling	え¯（柄）pattern	はし¯（端）edge	さくら¯（桜）	ともだち¯（友達）
1型 Type 1: fall at 1st	え＼（絵）painting	は＼し（箸）chopstick	か＼ぶと（兜）	お＼んせい（音声）
2型 Type 2: fall at 2nd	-	はし＼（橋）bridge	ここ＼ろ（心）	いろ＼がみ（色紙）
3型	-	-	おとこ＼（男）	みずう＼み（湖）
4型	-	-	-	いもうと＼（妹）

Excepting type 1, pitch starts with L.



Acoustic model
/ 音響モデル

Converts linguistic features to speech features

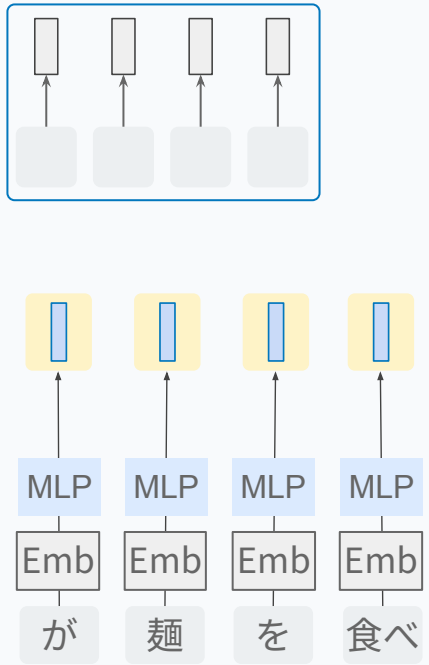


Encoder: how to extract features from input

Independent

✓ Streamable

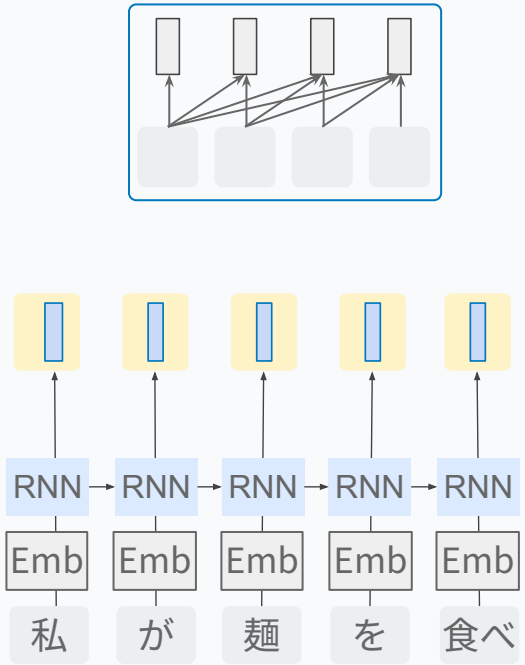
× No contexts from neighbors



Past input

✓ Streamable

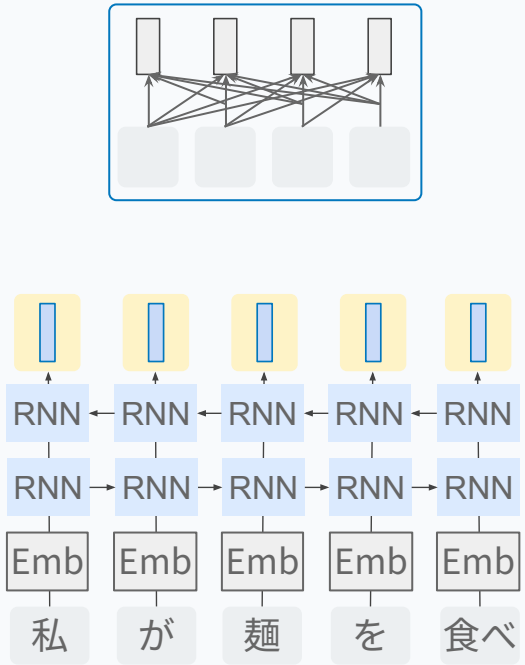
× Only past context



All inputs

✓ Context from all inputs

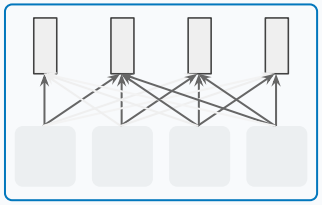
× non-streamable

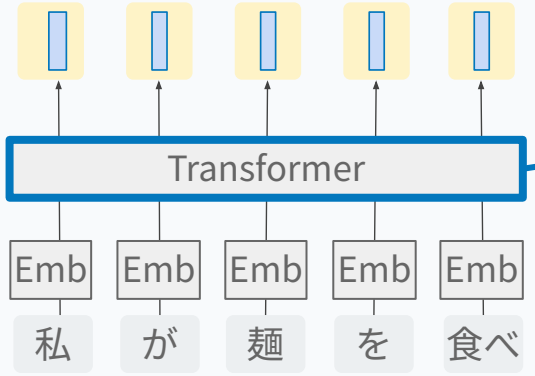


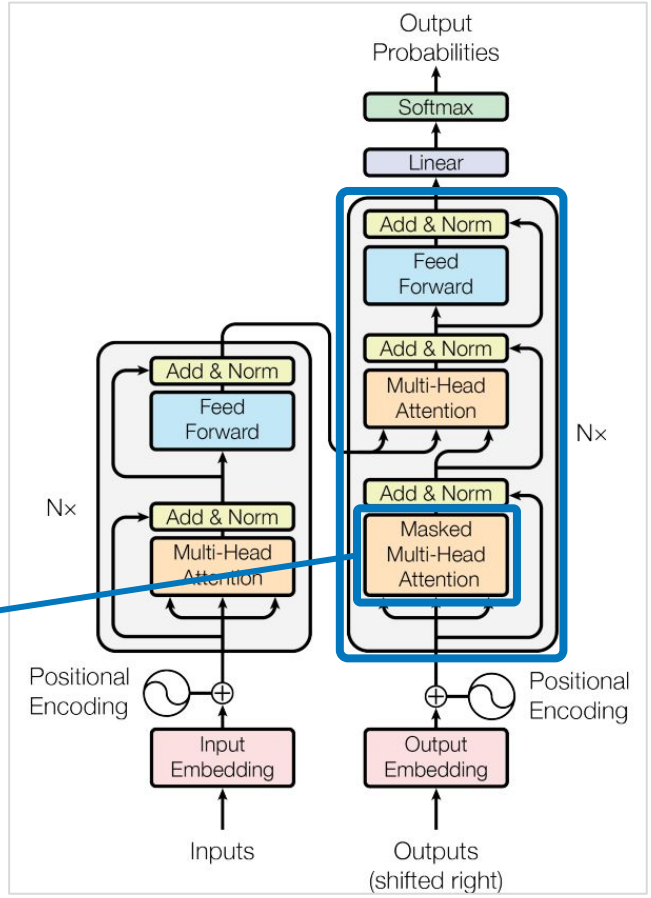
Transformer & attention mechanism

All inputs (with weights)

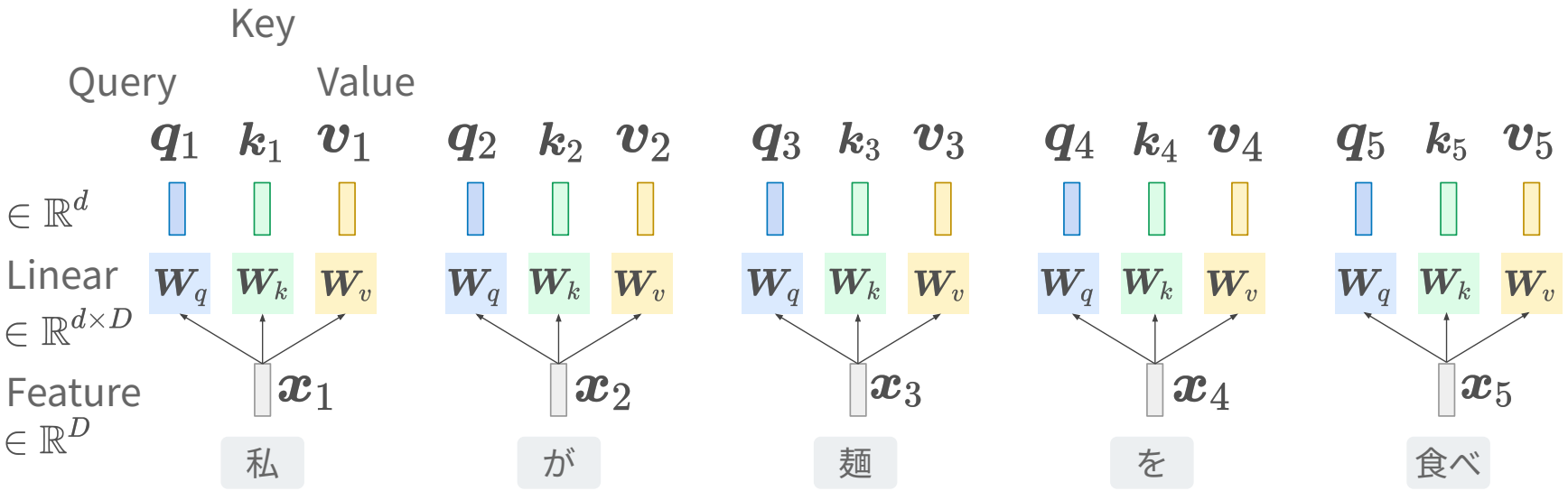
✓ long-contexts, learnable importance



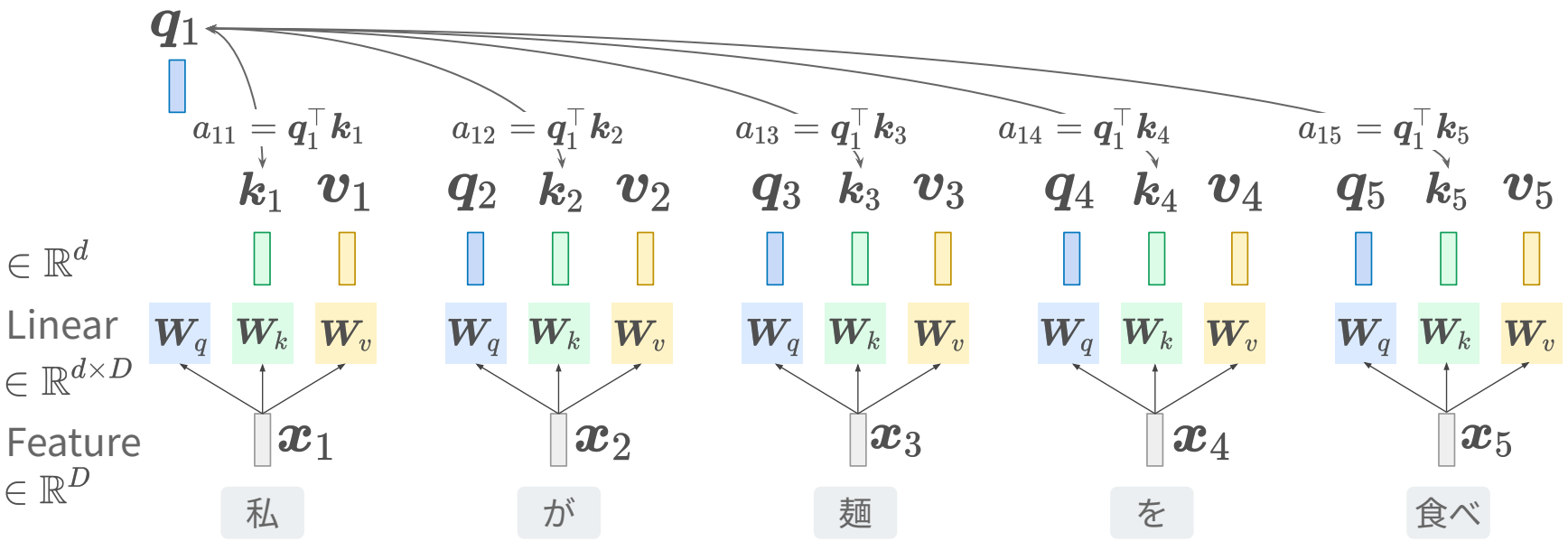




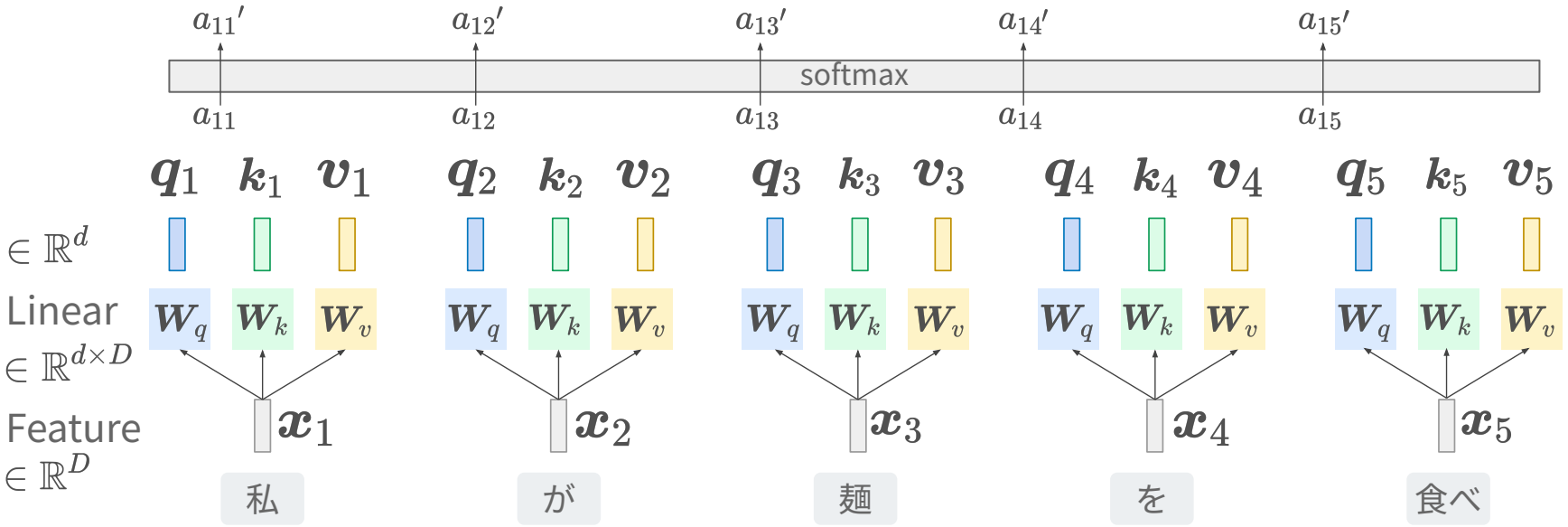
Self attention: sequence learning



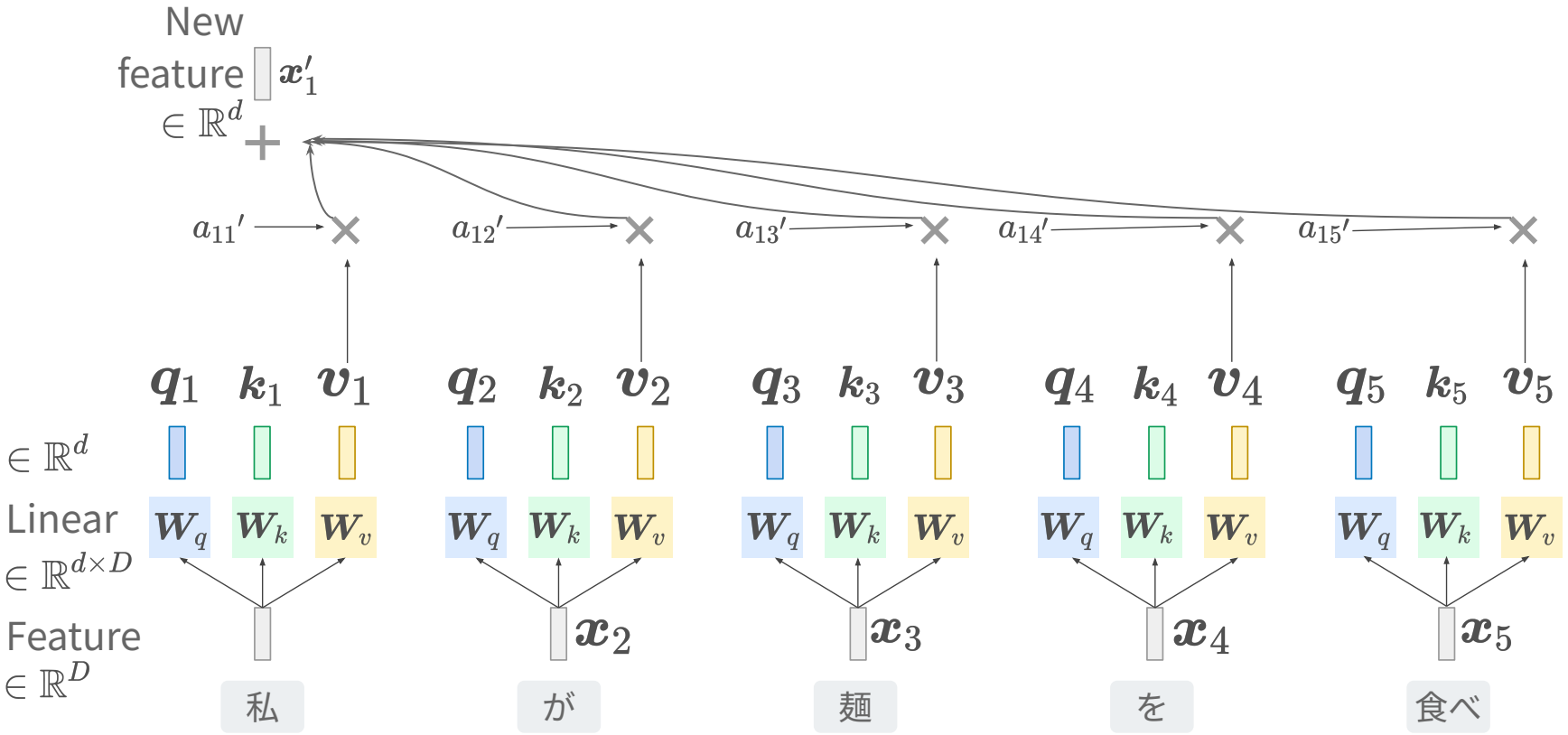
Inner product to calculate correspondence



Apply softmax to be “attention weights”



Weighted values become a new feature!



Further reading: NLP course for undegraduates

自然言語処理

慶應義塾大学 理工学部 情報工学科 講義

2026年度

事前知識の確認

- 第01回：イントロダクション
- 第02回：自然言語処理のための機械学習に入門する

言語とは何だろう

- 第03回：言語学に入門する

自然言語処理の基礎知識

- 第04回：単語をベクトルで表現する
- 第05回：言語モデルで文章を統計的に扱う
- 第06回：系列にラベリングする

これまでの復習

- 第07回：演習１

https://github.com/takamichi-lab/nlp-lecture-keio

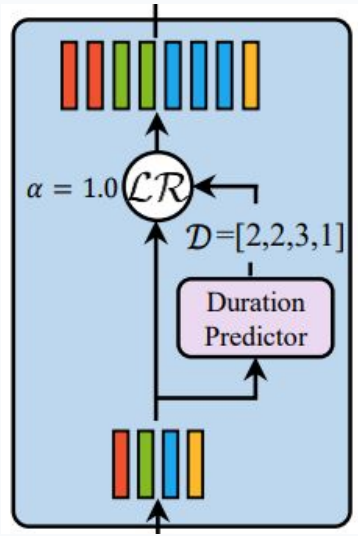
大規模言語モデル時代の技術

- 第08回：Transformer を理解する
- 第09回：言語モデルを事前学習する

Alignment

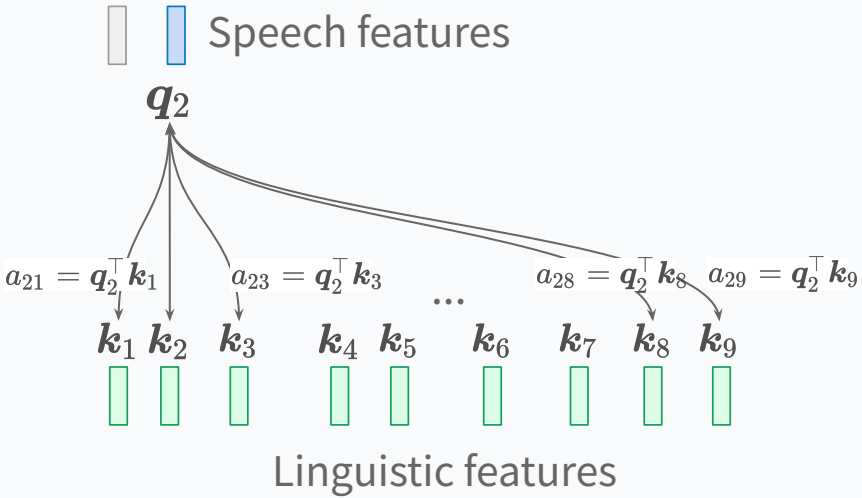
Hard alignment

Aligns one linguistic feat. to multiple speech feats, duplicating the linguistic feats. to match toe the speech feats.



Soft alignment

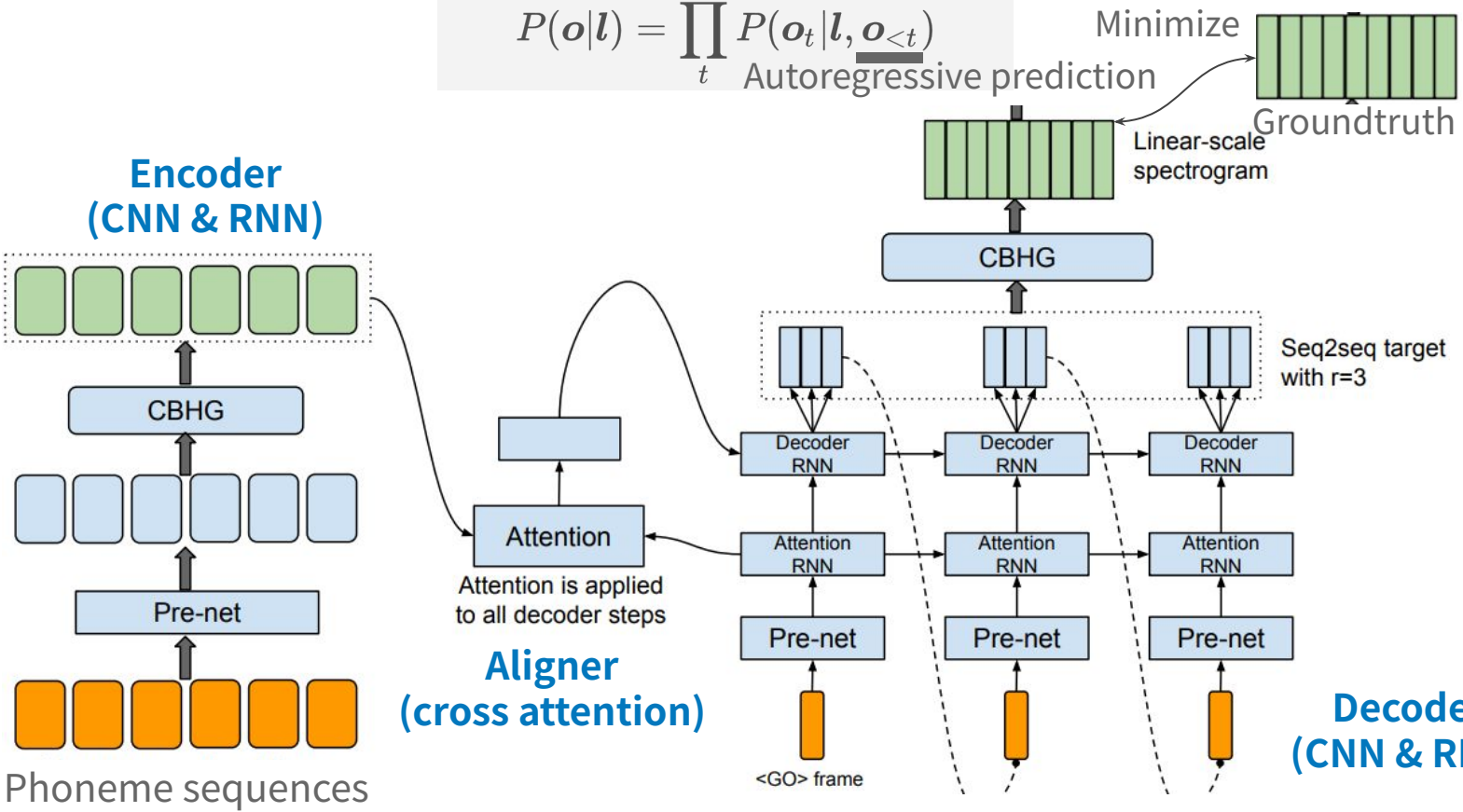
Uses all possible linguistic feats, e.g., cross-attention mechanism.



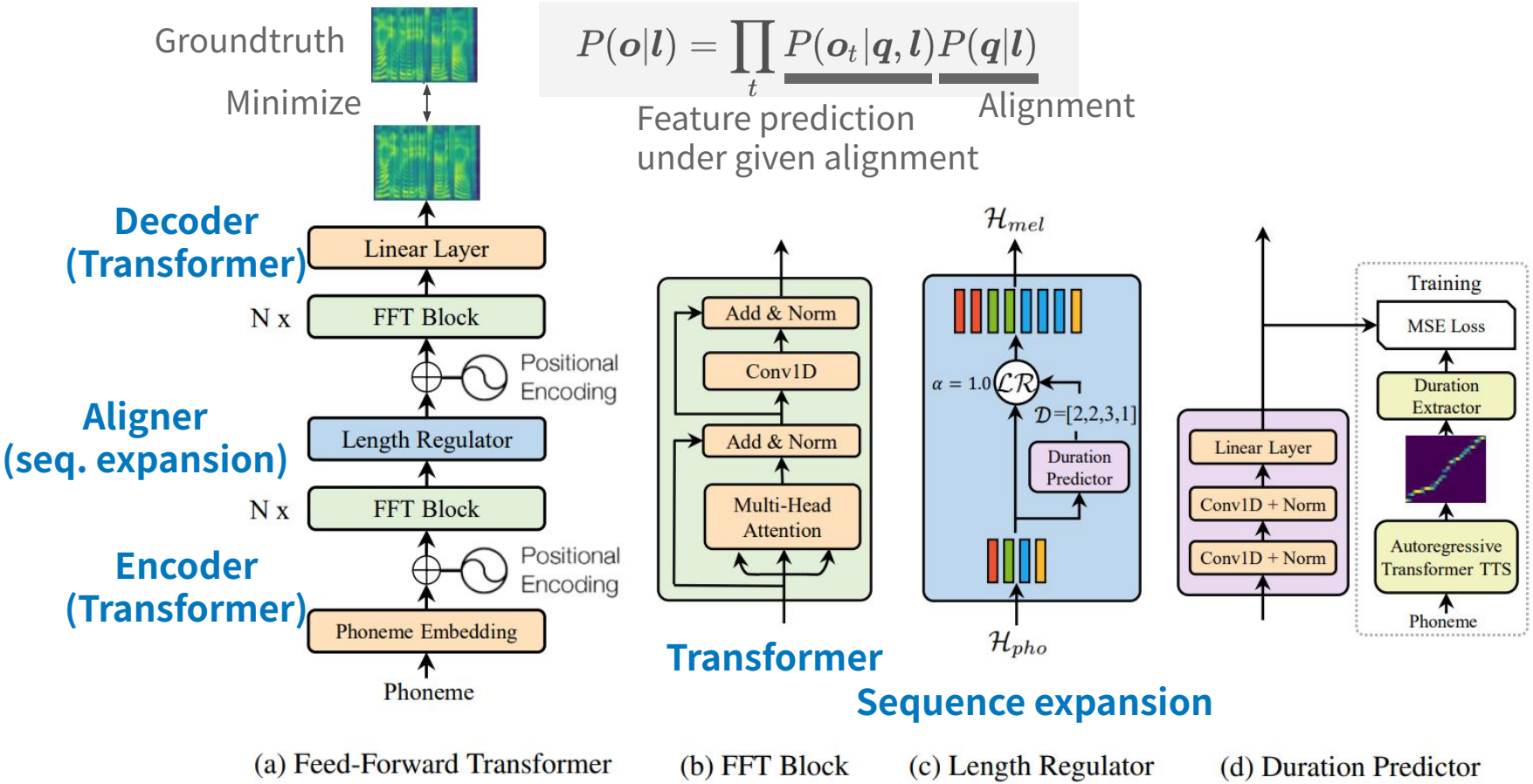
Tacotron: pioneer of end-to-end TTS (2017)

$$P(\boldsymbol{o}|\boldsymbol{l}) = \prod_t P(\boldsymbol{o}_t|\boldsymbol{l}, \boldsymbol{o}_{<t})$$

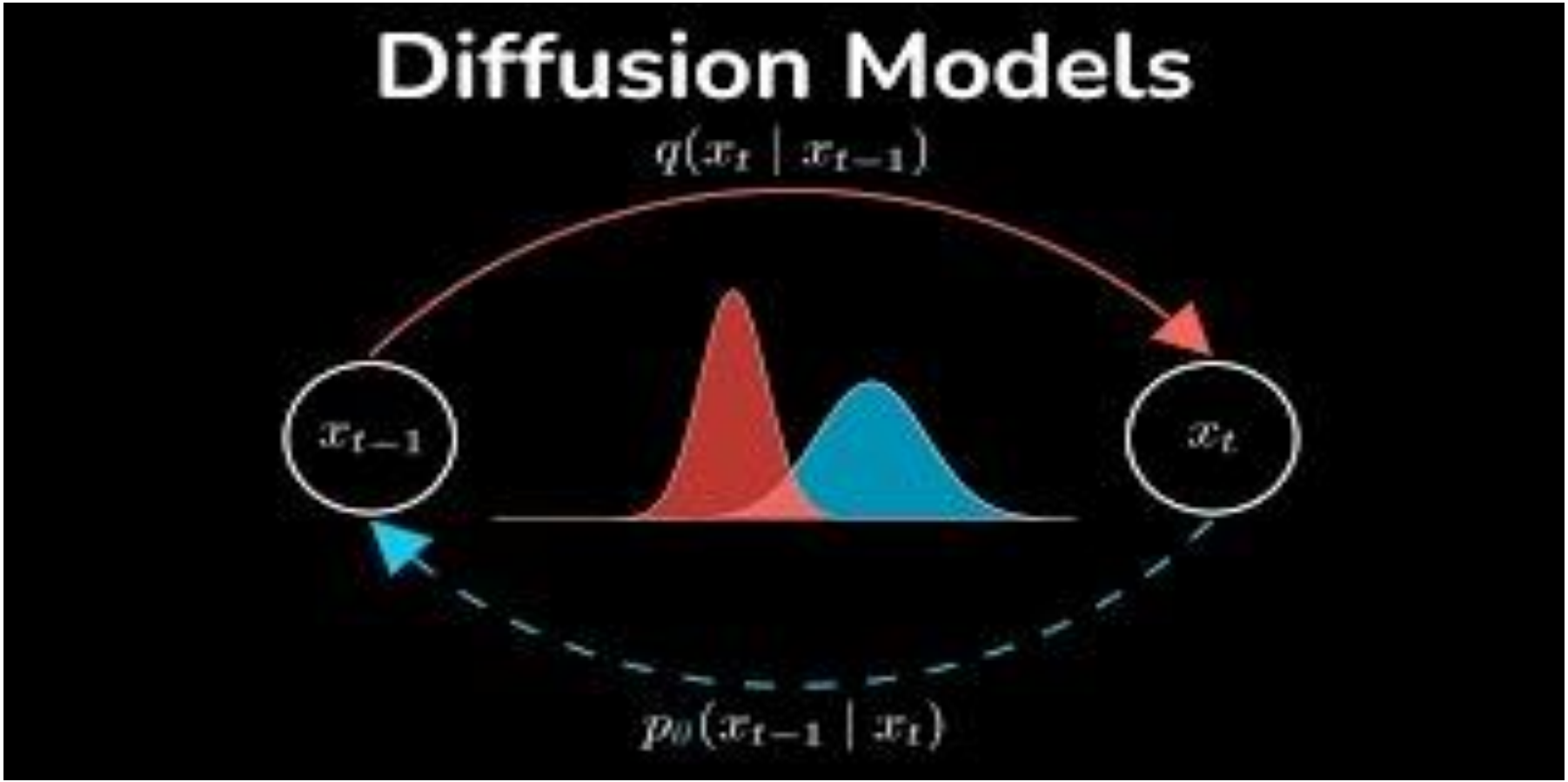
Autoregressive prediction



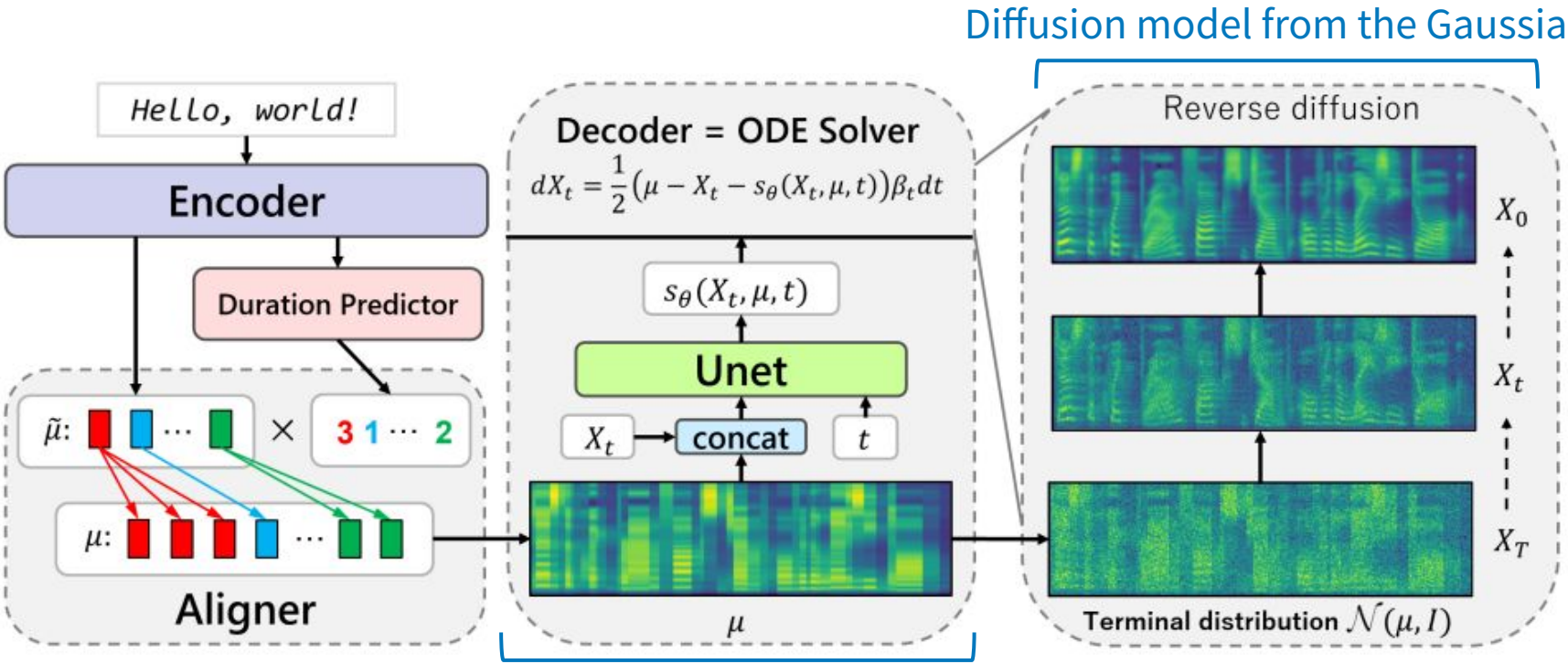
FastSpeech: duration-informed TTS



Diffusion model / 拡散モデル



Grad-TTS: first attempt to diffusion-based TTS

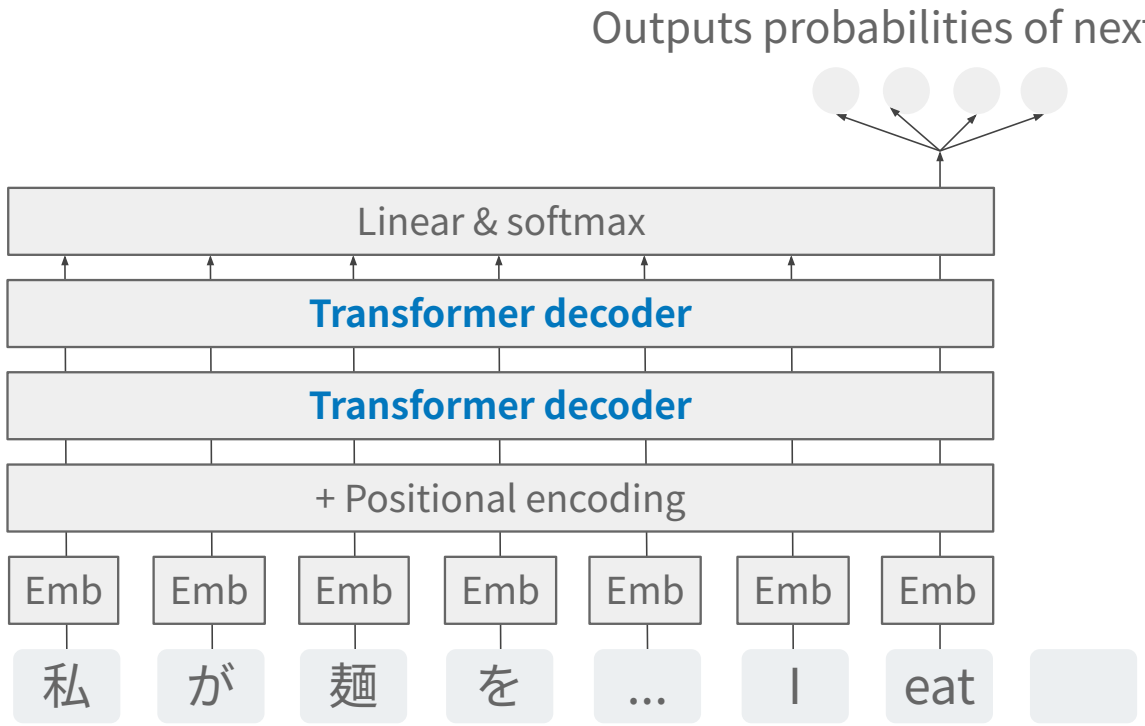


Advanced: optimal transportation and flow matching -> MatchaTTS

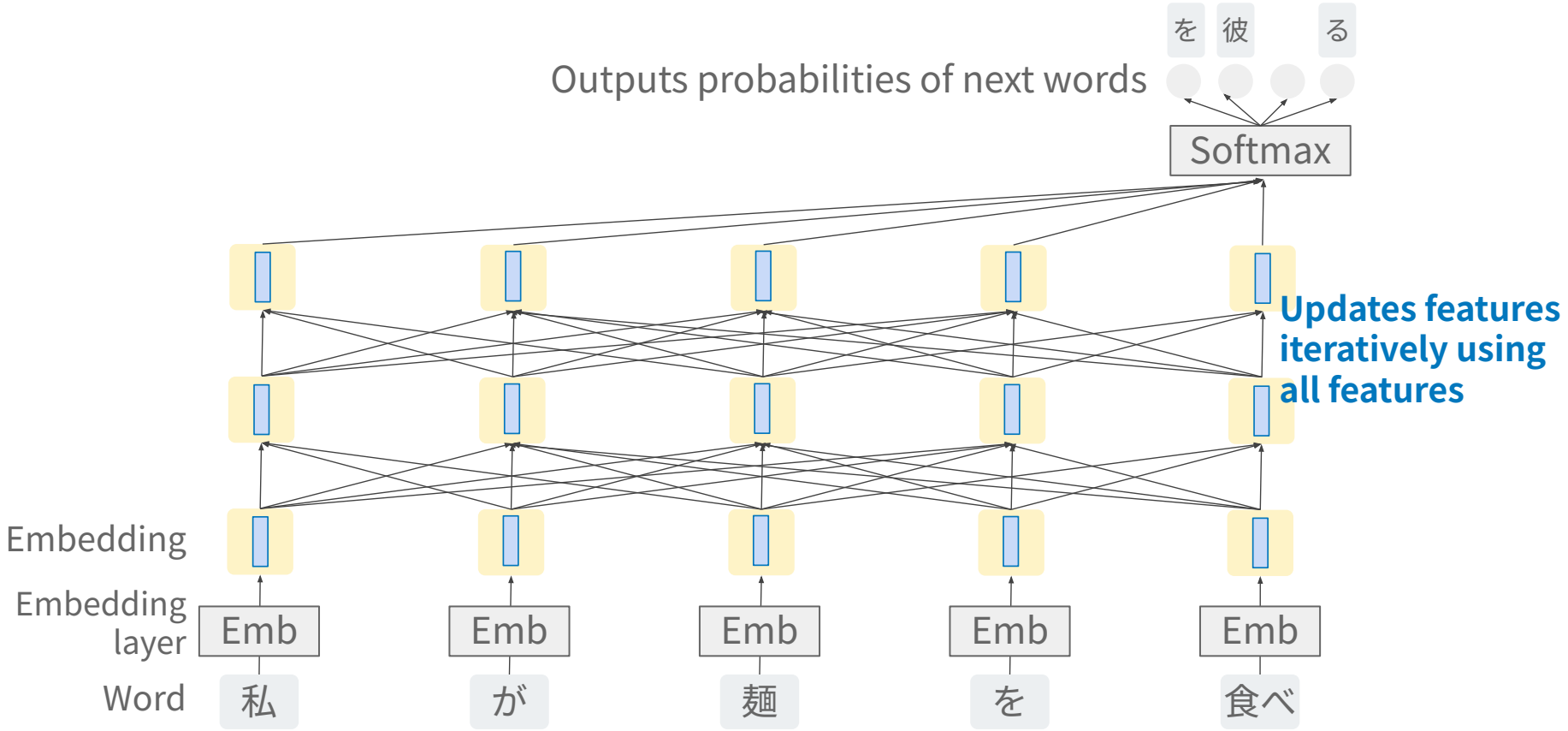
https://arxiv.org/pdf/2105.06337

LLM-powered TTS / LLMを用いた音声合成

LLM: large language model (Decoder-only model)

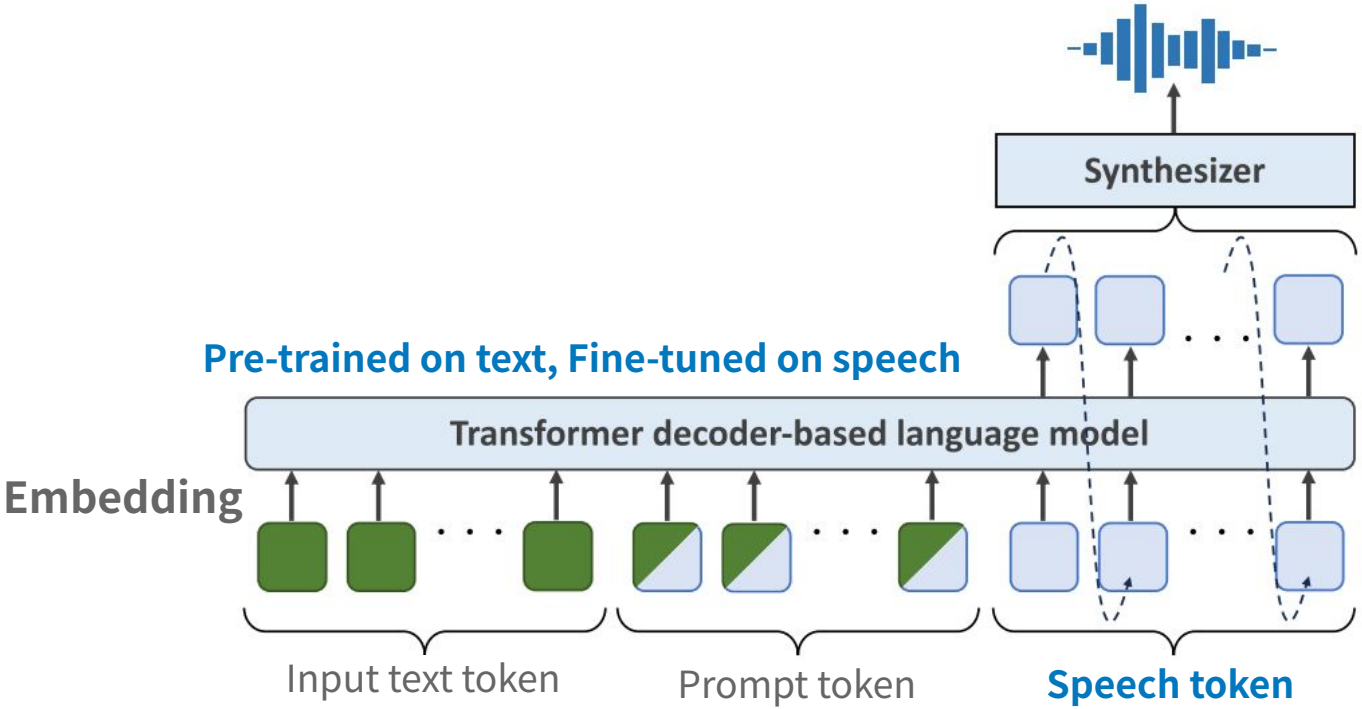


Concept of Decoder-only LLM



Typical LLM-based TTS

Predicts next speech token using a LLM.



https://www.jstage.jst.go.jp/article/jasj/81/10/81_719/_pdf/-char/ja

Why LLMs are used?

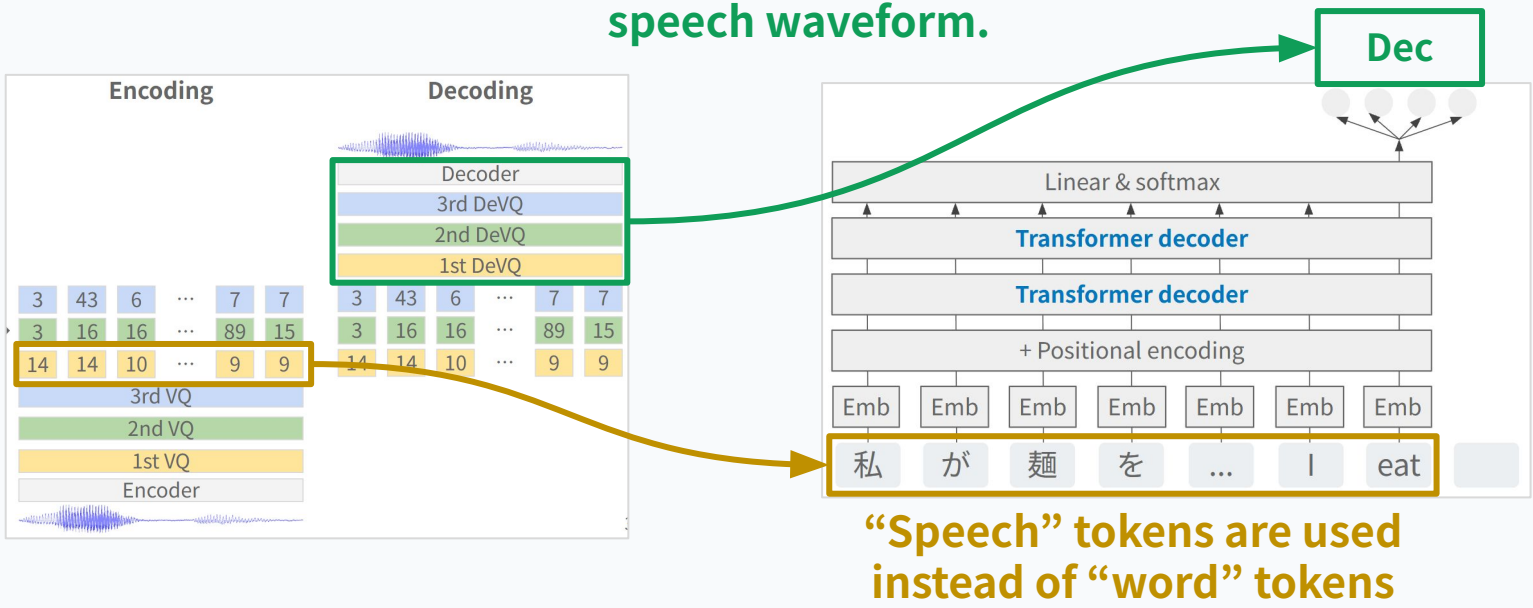
- **Large pre-training:** sequence modeling (feature extraction & score calculation) are pre-trained using text symbol sequences.
- **Language knowledge:** Pretrained models may acquire abilities of grapheme-to-phoneme conversion, accent estimation, etc.
- **Scalability:** Data size and model size (in log scale) affect to the performance linearly.

How to input/output speech to/from LLMs?

Discrete tokens

Speech-VQed symbols are used as like text symbols. Embedding layer is used for converting into continuous representations.

DeVQs convert output "speech" tokens to speech waveform.



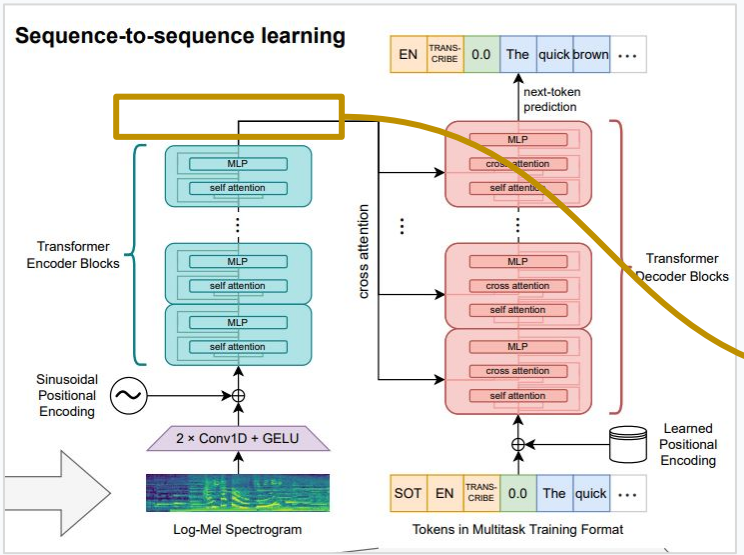
"Speech" tokens are used instead of "word" tokens

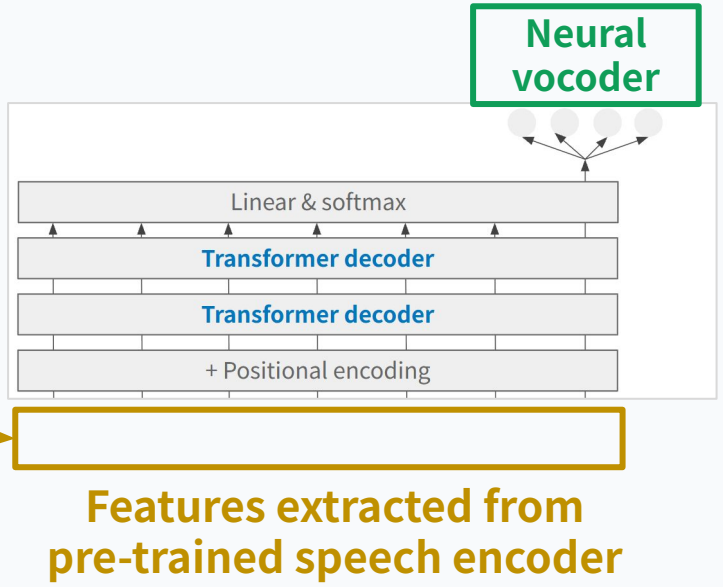
How to input/output speech to/from LLMs?

Continuous representations

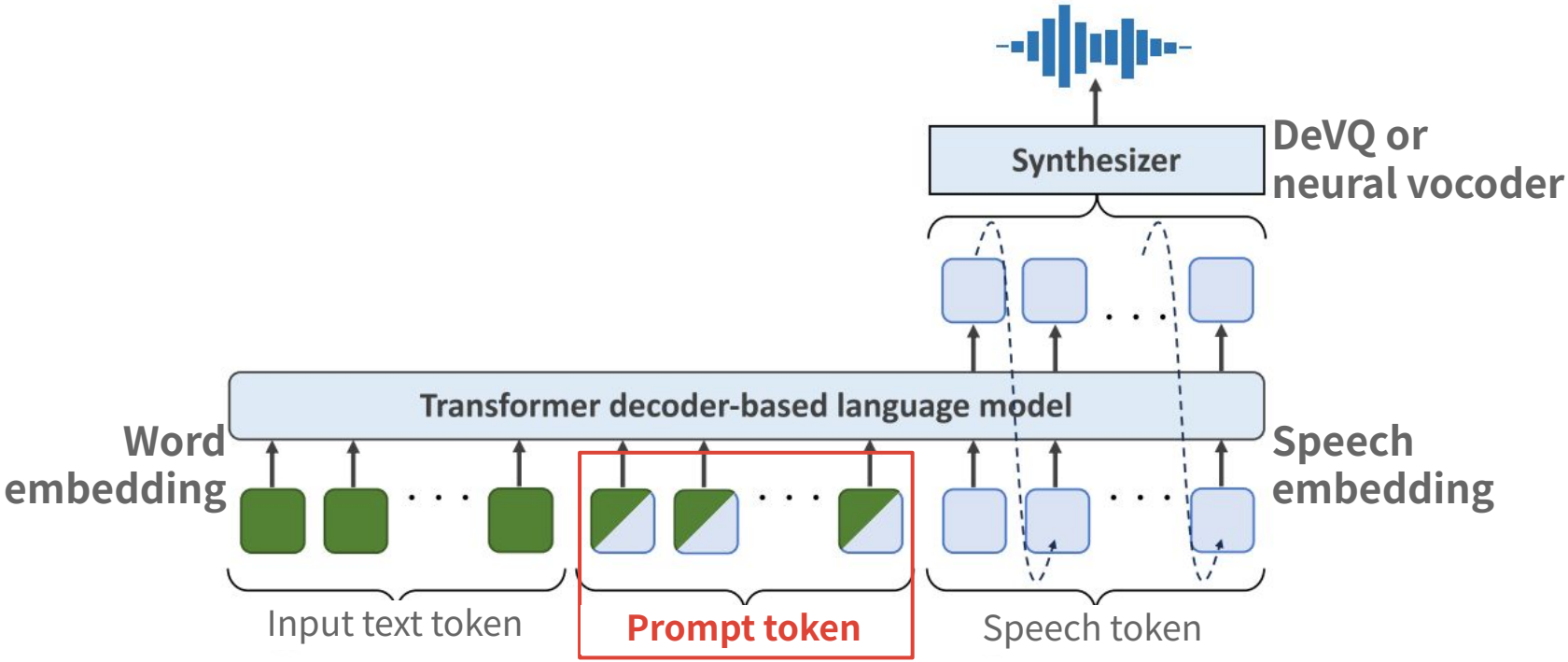
Continuous representations are directly used. Pre-trained speech encoders are used but embedding layer does not.

Whisper ASR models





Prompt



What is “prompt” in TTS?

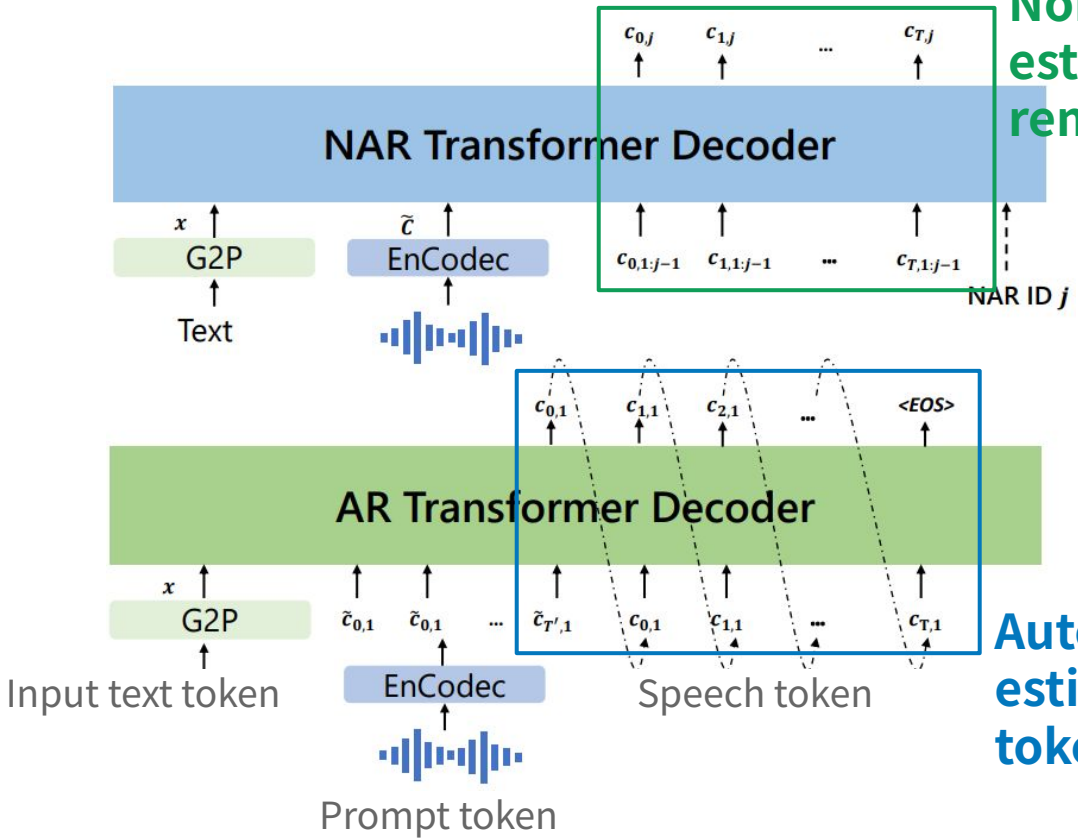
Speech prompt



Text prompt



VALL-E: frontier of LLM-based TTS



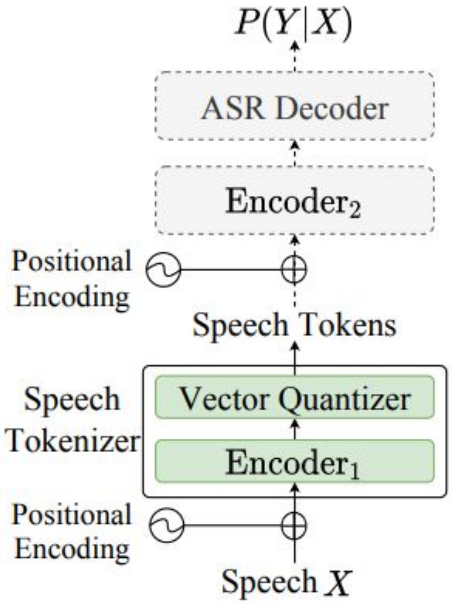
Non-autoregressive estimation of remaining-VQ tokens

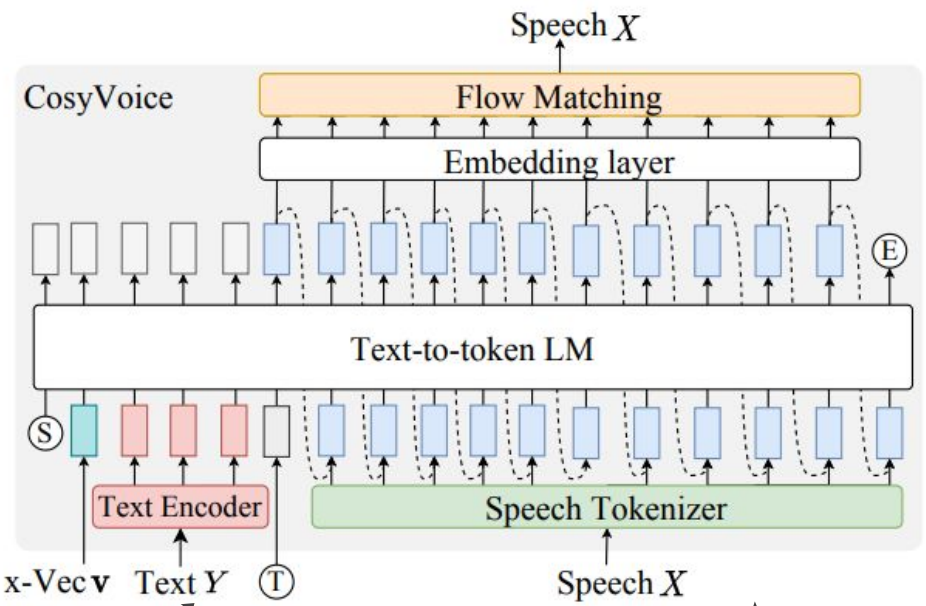
Autoregressive estimation of 1st-VQ tokens in RVQ

https://arxiv.org/abs/2301.02111

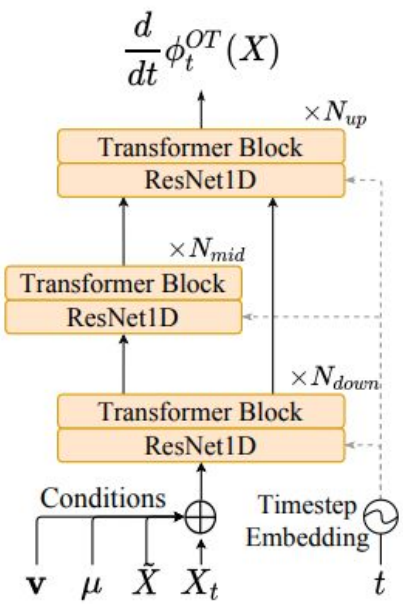
CosyVoice: flow matching & LLM-based TTS

Speech symbols are trained to predict text from speech



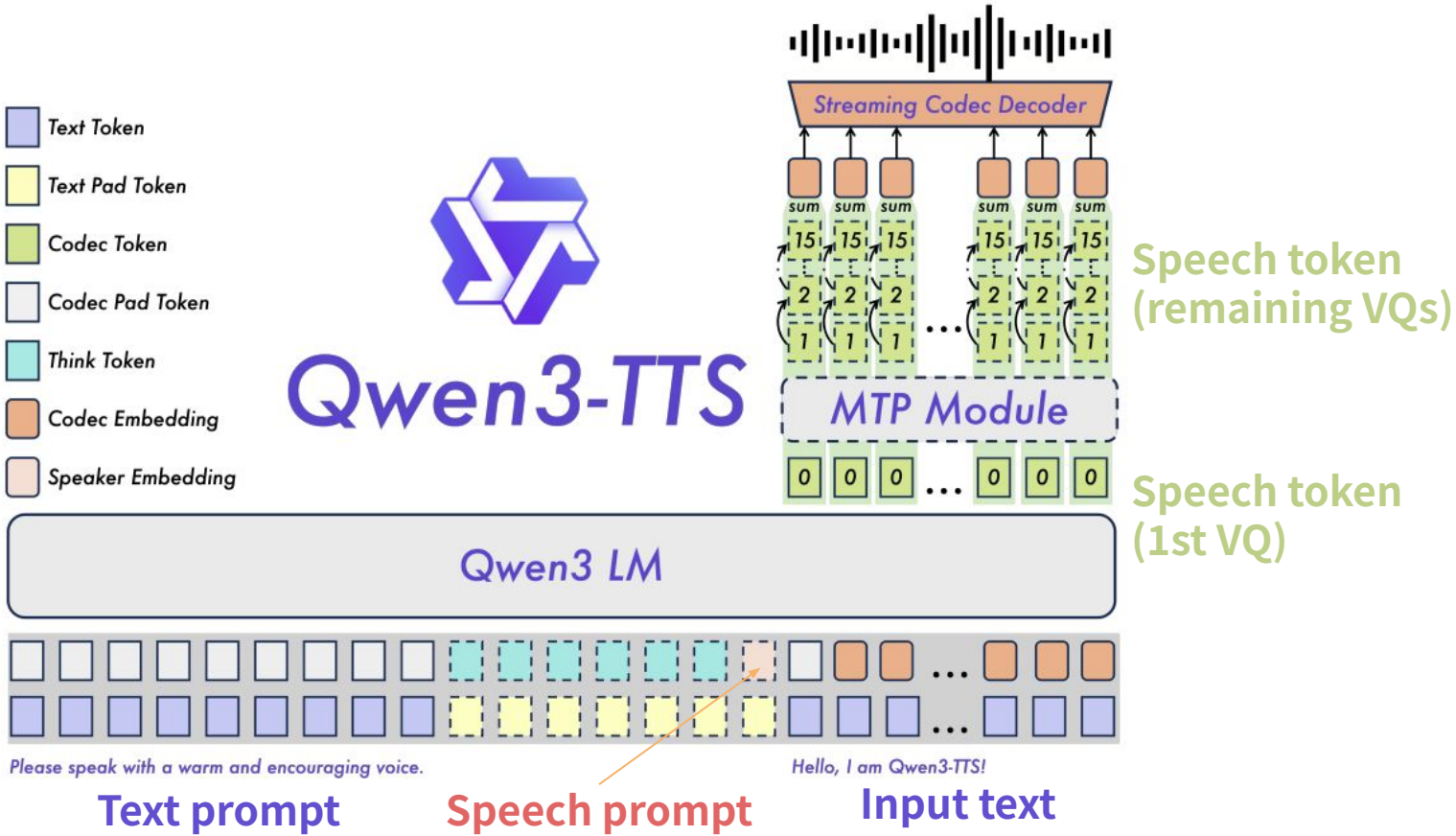


Flow matching-based decoder



Speech prompt **Input text token** **Speech token**

Qwen3-TTS: strong LLM backbone



Conclusion / まとめ

Conclusion

- History
  - 90 years of speech synthesis from electric circuits to computers

- Mathematical notation
  - TTS is one kind of sequence-to-sequence conversion

- Text analysis
- Acoustic models
- LLM-based